

SUHAAS CHOUDARY MALLAVARAPU

suhaaschoudarymallavarapu10@gmail.com · +91-9290770994 · [LinkedIn](#) · [GitHub](#) · [LeetCode](#) · [Portfolio](#)

SUMMARY

Computer Science sophomore with production deployment experience across full-stack MERN applications and ML-integrated systems. Built and shipped a cloud-deployed student management platform featuring JWT-secured REST APIs, RBAC, and MongoDB Atlas — and a CNN-based digit classifier with 98%+ validation accuracy. Pursuing a Software Engineering internship to contribute to backend systems, API architecture, and AI-driven product development.

EDUCATION

B.Tech — Computer Science & Engineering — SRM University AP | Aug 2024 – Present

CGPA: 7.4 · DSA, OOP, Operating Systems, DBMS, Computer Networks

Class XII (MPC) — Bhashyam Junior College | 2024

86%

Class X (ICSE) — Loyola Public School | 2022

90%

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), C, C++

Frontend: React.js, Tailwind CSS, HTML/CSS, Responsive Design

Backend: Node.js, Express.js, REST API Design, JWT Authentication, RBAC, Middleware

Databases: MongoDB Atlas, Mongoose ODM, PyMongo, NoSQL Schema Design

AI / ML: TensorFlow, Keras, NumPy, OpenCV, CNN Architecture, Streamlit

DevOps & Cloud: Git, GitHub, Render, MongoDB Atlas Cloud, Environment Config

Core CS: Data Structures & Algorithms, OOP, System Design, DBMS, OS

PROJECTS

EduTrack AI — Full-Stack MERN Student Management Platform *React.js · Node.js · Express.js · MongoDB Atlas · Mongoose · JWT · Tailwind CSS · Render*

GitHub: github.com/suhaas1126/EduTrack-AI Live: edutrack-ai-4.onrender.com

- ▶ Architected a production-grade MERN application with three distinct role-based dashboards (Admin, Teacher, Student), enforcing data isolation via JWT-secured routes and granular RBAC middleware.
- ▶ Designed and implemented 15+ RESTful API endpoints covering student records, attendance, grade management, and reporting — with centralized error handling and structured JSON response conventions.
- ▶ Configured MongoDB Atlas cloud database with Mongoose ODM; modeled normalized schemas for efficient multi-collection queries and persistent, scalable data storage.
- ▶ Deployed the full-stack application to Render with production environment variable management and cloud database connectivity; integrated analytics dashboards with AI-assisted academic risk flagging.

Handwritten Digit Classifier — CNN with Real-Time Inference *Python · TensorFlow/Keras · Streamlit · NumPy · OpenCV · MNIST*

GitHub: github.com/suhaas1126/CodeAlpha_HandwrittenRecognition

- ▶ Designed and trained a multi-layer CNN on 60K MNIST samples — Conv2D → MaxPooling → Dropout → Dense — achieving 98%+ validation accuracy with optimized regularization to minimize overfitting.
- ▶ Built a custom preprocessing pipeline to normalize pixel values, reshape tensors, and invert freehand canvas inputs to match MNIST distribution, bridging the domain gap between sketched and training data.
- ▶ Deployed an interactive Streamlit inference app with a live drawing canvas, delivering sub-second predictions and per-class confidence scores; structured codebase into independent training, inference, and UI modules.

CERTIFICATIONS & ACTIVITIES

MongoDB Associate Developer (2024) · Verified expertise in NoSQL data modeling, aggregation pipelines, indexing strategies, and CRUD operations across 5 assessment domains.

Tech Education Channel (50+ Videos) · Produces long-form engineering content on emerging technologies; demonstrates technical depth, independent research, and communication skills.